

# Lithium-Lanthanum-Zirconate Solid Electrolytes for High-Density Energy Storage: A Comparative Study

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## Abstract

This study investigates the ionic conductivity and electrochemical stability of garnet-type lithium-lanthanum-zirconate (LLZO) solid electrolytes as a candidate material for next-generation solid-state lithium batteries. We report a room-temperature ionic conductivity of  $1.2 \times 10^{-3}$  S/cm for aluminum-doped LLZO samples sintered at 1230°C, representing a 35% improvement over previously reported undoped LLZO formulations.

## 1. Introduction

Conventional lithium-ion batteries rely on liquid organic electrolytes, which pose safety risks including flammability and electrolyte leakage. Solid-state electrolytes have emerged as a promising alternative, offering improved thermal stability and the potential for higher energy density through compatibility with lithium metal anodes.

Among solid electrolyte candidates, garnet-type LLZO has attracted considerable attention due to its high ionic conductivity, wide electrochemical stability window, and chemical compatibility with lithium metal. However, challenges remain in achieving consistent grain boundary conductivity and minimizing interfacial resistance with electrode materials.

## 2. Methodology

Samples of aluminum-doped LLZO ( $\text{Li}_{6.25}\text{Al}_{0.25}\text{La}_3\text{Zr}_2\text{O}_{12}$ ) were synthesized via solid-state reaction using high-purity precursor powders. The powders were calcined at 950°C for 6 hours, ball-milled, and subsequently sintered at temperatures ranging from 1150°C to 1280°C for 8 hours under an argon atmosphere to minimize lithium volatilization.

Ionic conductivity was measured using electrochemical impedance spectroscopy (EIS) across a frequency range of 1 Hz to 1 MHz at temperatures from 25°C to 100°C. Sample density was determined using the Archimedes method, and phase purity was confirmed via X-ray diffraction (XRD).

### 3. Results

Samples sintered at 1230°C exhibited the highest relative density at 96.4%, correlating with the highest measured ionic conductivity of  $1.2 \times 10^{-3}$  S/cm at room temperature. Samples sintered below 1180°C showed significantly reduced density (below 88%) and correspondingly lower conductivity, attributed to incomplete densification and increased grain boundary resistance.

The activation energy for lithium-ion migration, calculated from Arrhenius plots, was found to be 0.31 eV for the optimally sintered samples, consistent with values reported for high-performance garnet electrolytes in prior literature. Cyclic voltammetry confirmed an electrochemical stability window extending to approximately 6 V relative to Li/Li<sup>+</sup>, indicating compatibility with high-voltage cathode materials.

### 4. Discussion

The observed improvement in ionic conductivity at the 1230°C sintering condition is attributed to optimal grain growth and reduced porosity, which together minimize obstruction of lithium-ion transport pathways across grain boundaries. These findings suggest that sintering temperature is a critical and highly sensitive parameter in optimizing LLZO performance for practical solid-state battery applications.

## 5. Conclusion

Aluminum-doped LLZO sintered at 1230°C demonstrates ionic conductivity and electrochemical stability sufficient for consideration in next-generation solid-state lithium battery designs. Future work should focus on reducing interfacial resistance between the LLZO electrolyte and lithium metal anodes, as well as scaling synthesis methods for industrial production.